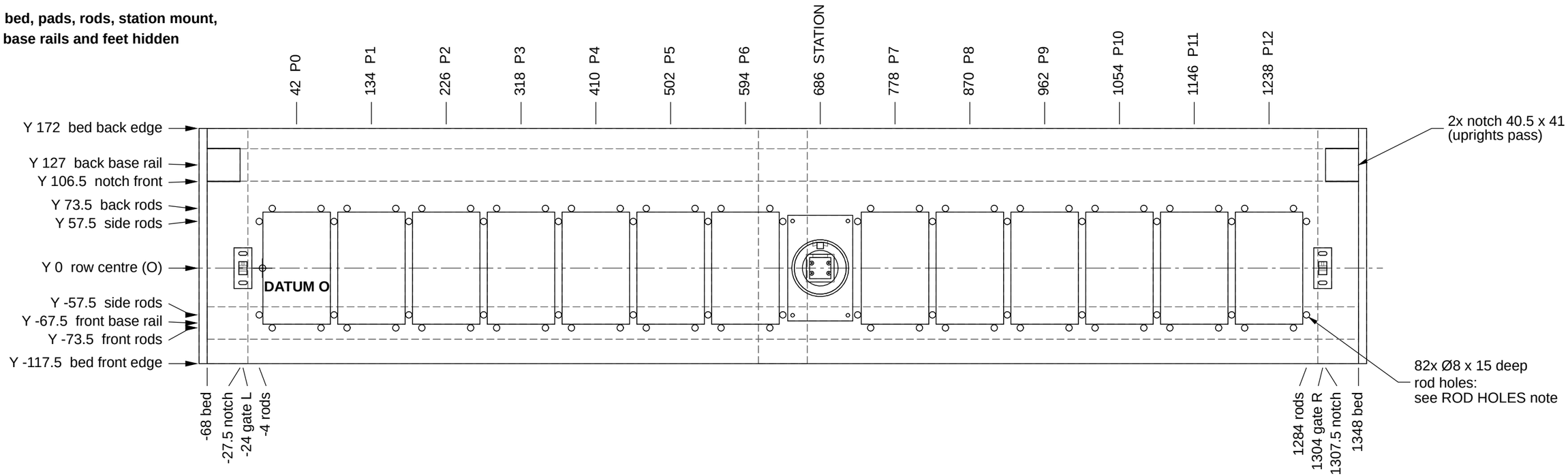
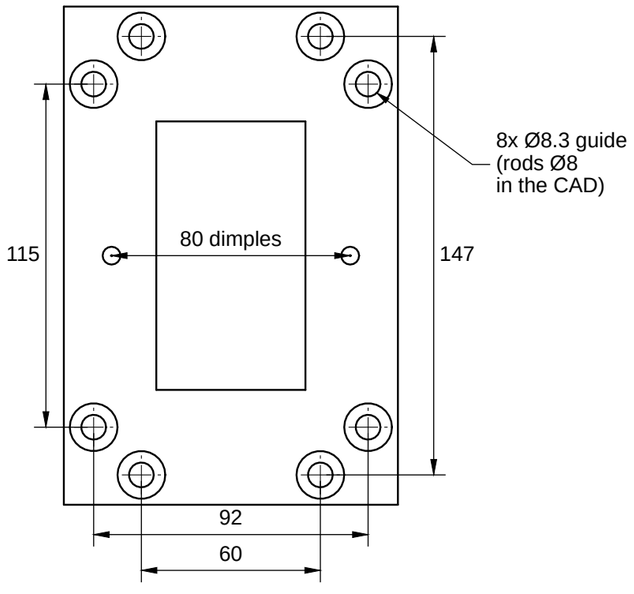
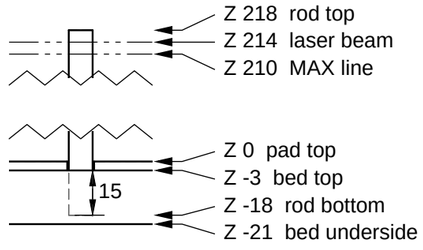
PLAN (1:5): bed, pads, rods, station mount, gate posts; base rails and feet hidden



DETAIL C (1:2): one cell's rod holes = drill_template
(cell centre at 0, 0; the machine drills them, BUILD_GUIDE 5.6)



DETAIL D (1:2): rod in the bed,
front, broken



Bed and pads

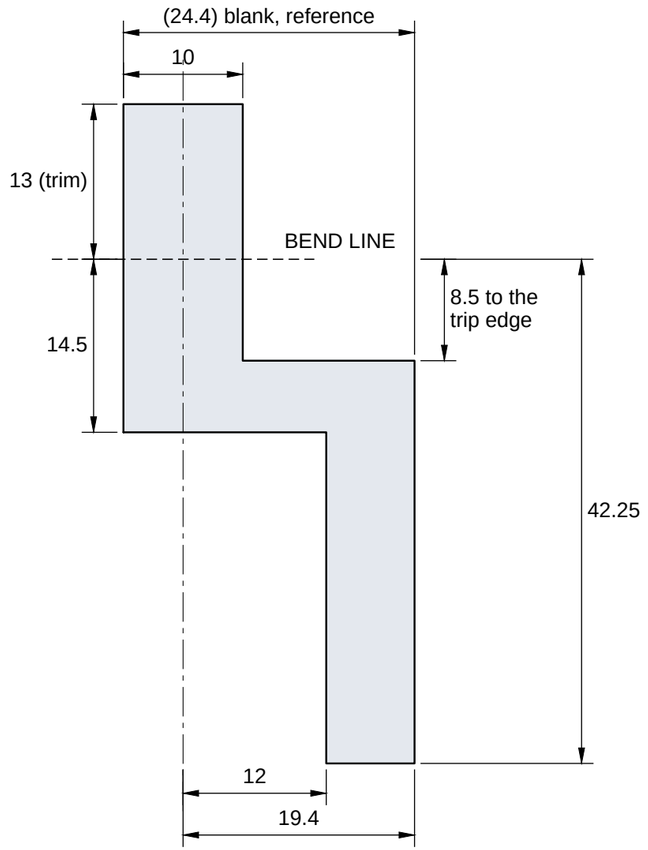
Item	Size (from the CAD)	Note
Bed, plywood (W)	1416 x 289.5 x 18	cut on site; test-drill for voids first (BUILD_GUIDE 5.1)
Pads (EVA + microfiber), 13	83 x 138 x 3	spray adhesive, after the rods (5.9)
Rods (dowels), 82	D-02 item C-11	glued; plumb within 1 over 200 [Guessing target] (5.7)

Datum O = machine origin (X 0; Y 0 = the pocket-row centre line), as the software uses it; X ordinates from O. The machine drills the rod holes through drill_template, so they are not laid out from the bed edges (BUILD_GUIDE 5.6). Station mount: cell 7, four #8 wood screws, no pad. Gate posts: bare plywood, no shim (5.9).

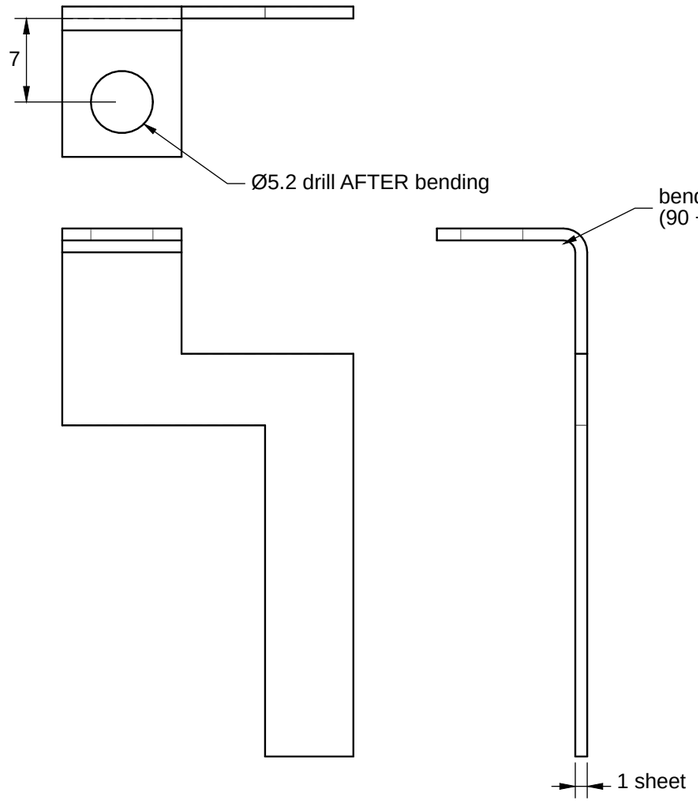
ROD HOLES: 82x Ø8 (ROD_D; brad-point bit to suit the dowels: 5/16 in for the BOM's 7.94 dowels) x 15 deep (ROD_EMBED), square to the bed (guide block, 5.6). Side rods on the wall lines: X -4, then every 92, at Y ±57.5. Front/back rods: pairs 60 apart about each pocket centre, at Y ±73.5; none in cell 7.

SlabRail - graded-slab batch scanner	**SHEET D-03 of D-17**
Bed, pockets and guide rods	SCALE 1:2, 1:5 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ 5fe0adc (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

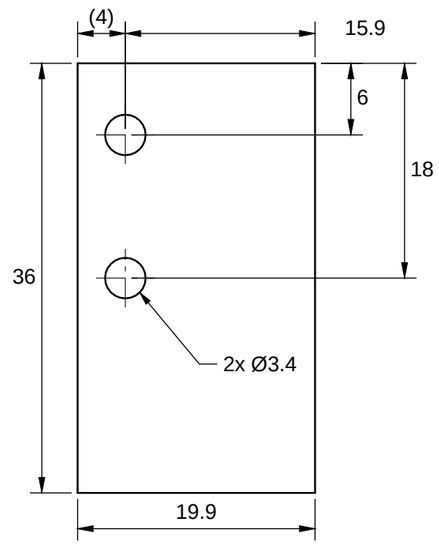
PROBE FLAG BLANK (2:1), flat, before bending
u from the neck centre line (= hole line), v from the bend line



PROBE FLAG AS BENT (2:1), for reference
(head frame: X along the row, Z above the cup face)



CARRY FLAG (2:1), flat
(seen from the front; top edge = trip edge)



Probe flag (BUILD_GUIDE 7.6; build.py probe_flag_blank.svg notes)

Aluminium sheet, 1 thick (cut_list C-07). 1. Cut and file to the outline; the blade's top edge is the trip edge: straight, square. 2. This view shows the face toward the buffer. Neck in the vise, jaws at the BEND line; bend the tab 90° TOWARD YOU. 3. THEN drill on the centre line, 7 from the plate's inner face. 4. Trim the tab to 2 past the hole. 5. Deburr: it rides in head_foot's guide slot and the sensor slot. Outline: cad/out/probe_flag_blank.dxf (1:1 print: .svg).

Carry flag (BUILD_GUIDE 7.4a)

Aluminium sheet, 1 thick, flat (cut_list C-10). File the top edge straight. Fix to carry_flag_arm's leg end with 2 x M3x8, inner edge toward the spine, top edge flush with the arm's top.

Tolerance: outlines PROPOSED +/-0.2 (no doc states one); bend 90 +/-0.5° (BUILD_GUIDE 7.6). The trip points are set by adjustment on assembly (probe 7.6, carry 7.4a), not by these outlines.

SlabRail - graded-slab batch scanner	**SHEET D-06 of D-17**
Flat parts (probe flag, carry flag)	SCALE 2:1
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ 5fe0adc (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**